

Session-05: Binomial Distribution

1. Find the probability of getting a total of 7 at least once in three tosses of a fair dice.

Ans: $91/216$

2. If 20% of the bolts produced by a machine are defective, determine the probability that out of 4 bolts chosen at random (i) 1, (ii) 0, (iii) less than 2 bolts will be defective.

Ans: (i) 0.4096

(ii) 0.4096

(iii) 0.8192

3. A communication system consists of 'n' components, each of which will, independently function with probability 'p'. the total system will be able to operate effectively if at least one-half of its components function. For what values of 'p' is a 5-component system more likely to operate effectively than a 3-component system?

Ans: $p > 1/2$